Loan Fraud detection project proposal

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1 Introduction

Loan Fraud is a significant concern for financial institutions, leading to substantial financial losses and reputation damage. As fraudsters become more sophisticated, there is a pressing need for a reliable and efficient system to detect and prevent fraudulent activities in loan applications.

2 problem statement

Superlender, a local digital lending company, leverages credit models to deliver profitable, high-impact loan alternatives.
Their approach focuses on two key drivers of loan default: willingness to pay and ability to pay. This project proposes the development of a Loan Fraud Detection system that leverages artificial intelligence to identify suspicious patterns and improve decision-making in loan processing.

3 Objectives

1. To develop a predictive model that accurately determines each application's loan repayment likelihood.
2. To enhance the efficiency of loan processing by automating the detection of fraudulent activities
3. To reduce financial losses and reputational risks for financial institutions.
4. To strengthen trust between lenders and customers by ensuring fair and secure lending practices.

4 Data Description

The data is approximately 14,743 rows, 28 columns and some columns contain nulls, duplicates.Test data was also provided for this challenge.

1. Demographics Data: (traindemographics.csv) Captures information personal about the borrower, such as age, employment status, location and education level which will enable us profile the borrower and their financial stability.

2. Previous Loans Data: (trainprevloans.csv) Provides historical information about loans previously accessed by the customer. The fields include loan amount,firstrepaiddate. This data provides insight into the repayment behaviour and creditworthiness.

3. Performance Data:(trainperf.csv) Focuses on the specific loan application from which you need to predict repayment.
The data fields provided include,loan amount, target variable(`good_bad_flag`).

5 Proposed approach

- Data Processing Data cleaning will handle missing values and outliers. Exploratory Data Analysis (EDA) understands feature and target relationships. Feature engineering creates new features to improve model performance. Feature scaling ensures unbiased features.

- Model Selection and Training

  We'll choose classification algorithms (Logistic Regression, Random Forest, SVM, Gradient Boosting), tune their hyperparameters, and train them on a portion of the data to learn feature-loan outcome relationships.

- Model Evaluation

  Divide data into training and validation sets to prevent overfitting. Train models on training set and evaluate on validation set. Calculate misclassification rate on validation set for accuracy assessment.
  Compare performance of different models and select the one with lowest misclassification rate.

- Model Deployment and Refinement

  Final Model Selection: Choose the best-performing model from the validation stage. Deployment: The final model can be integrated into SuperLender's loan application process to predict repayment odds for new loan applications. Monitoring and Improvement: Monitor the model's performance over time and retrain it periodically with new data to maintain accuracy.

6 Deliverables

A loan fraud detection system that leverages artificial intelligence to identify suspicious patterns and improve decision-making in loan processing.

7 Timelines(6 weeks)

Activity	Weeks 1	Weeks 2	Weeks 3	Weeks 4	Weeks 5	Weeks 6
Data collection, preprocessing, feature engineering.						
Model development and evaluation.						
Deployment, documentation, testing.						
Outcome: Loan Fraud Detection system for SuperLender.						

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